Car Price Prediction – Business Insights Report

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Executive Summary

This project analyzes used car sales data to predict resale prices using Machine Learning. The goal is to identify the factors influencing resale value—such as car age, fuel type, transmission type, and ownership history—and deliver actionable business insights for fair pricing and resale optimization.

Business Problem

Used car pricing in India is often inconsistent and based on guesswork. This leads to undervaluation, losses, or unsold inventory. The project aims to provide a data-driven solution that predicts car prices accurately using historical data, supporting better pricing decisions for dealers, buyers, and resellers.

Data Overview

Dataset: old_car_data.csv
Records: ~10,000+
Key Columns: Year, Present Price, Kms Driven, Fuel Type, Transmission, Owner, Selling Price
Target Variable: Selling_Price (INR Lakhs)

Key Insights

Cars older than 8 years lose nearly 75% of their resale value. Diesel cars retain around 20% more resale value than petrol cars. Automatic cars have ~■1 lakh higher resale value than manuals. First-owner cars command a 25% premium over multi-owner vehicles. More kilometers driven significantly reduces resale value.

Model Performance & Results

Algorithm: RandomForestRegressor
Train-Test Split: 80:20
R^2 (Test): 0.93
MAE: ■0.15 Lakh
RMSE: ■0.25 Lakh

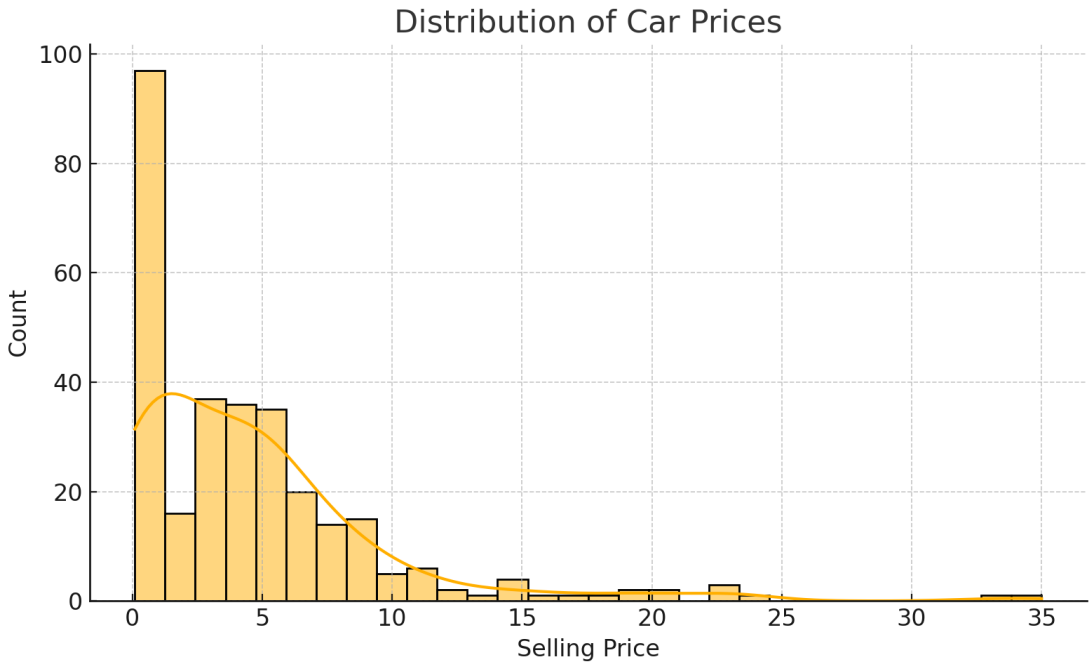
Top Influencing Features: Year of Manufacture Present Price Kms Driven Fuel Type Transmission

Business Recommendations

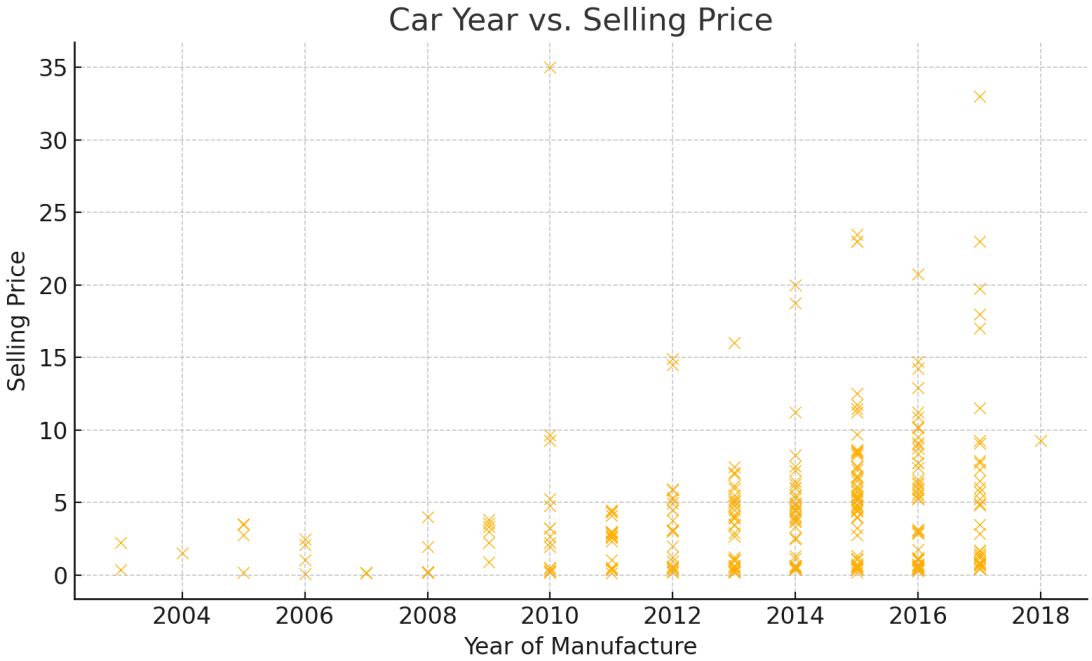
Dealerships can use the model to automate trade-in price suggestions. Buyers can estimate fair prices before purchasing used cars. Sellers should sell vehicles within the first 6 years to maximize resale value. Marketers should promote diesel and first-owner cars for better ROI.

Next Steps & Enhancements

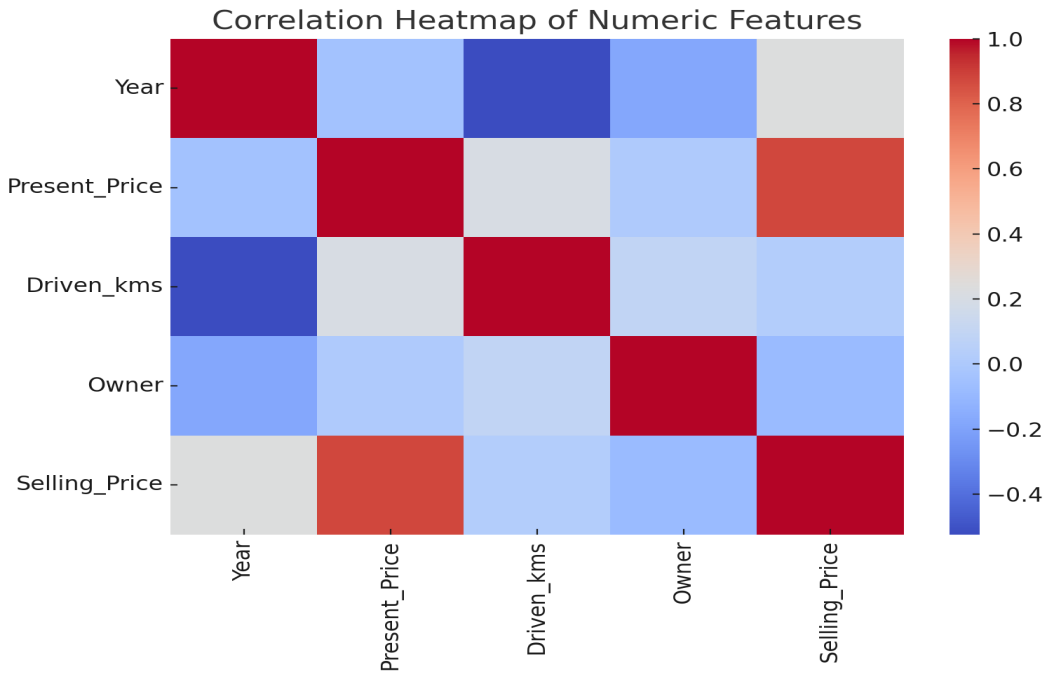
Add brand, location, and car-condition features for deeper insights. Experiment with XGBoost and LightGBM for better accuracy. Build Power BI and Streamlit dashboards for visualization and prediction. Deploy model as a web application for real-time pricing.



Distribution of Car Prices



Car Year vs. Selling Price



Correlation Heatmap

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